

State-Dependent Hedging: Swap Spreads and Credit Exposure

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Abstract

This paper studies whether swap spread positions hedge credit exposure and whether that hedging effectiveness depends on the macroeconomic state. Swap spreads—the difference between fixed rates on interest rate swaps and Treasury yields of similar maturity—are widely used in relative-value and risk-management strategies. Practitioners often view receive-fixed swap positions combined with long Treasury holdings as providing protection during credit sell-offs, yet episodes driven by inflation or monetary tightening suggest that this hedge can fail. We develop a parsimonious intermediary-based asset pricing framework in which balance sheet constraints and safe-asset demand drive time variation in swap spreads and credit risk premia. The model yields state-dependent predictions: swap spread positions hedge credit exposure under funding-stress shocks but lose hedging power under inflation or rate-driven shocks. We then evaluate these predictions using swap-spread-replicating portfolios across maturities and standard measures of credit exposure, documenting strong hedge performance in funding-stress episodes and weak or adverse performance in inflation-driven drawdowns. The findings link swap spread dynamics to intermediary constraints and provide guidance for state-contingent risk management.

Keywords: swap spreads; credit spreads; hedging; intermediary constraints; flight-to-quality; state dependence

JEL Codes: G12; G13; G21; E44

1 Introduction

Financial markets rely heavily on hedging relationships that are assumed to provide protection in times of stress. Yet, history repeatedly shows that hedges that work in some downturns fail in others. This paper studies one such relationship: whether positions in swap spreads hedge credit exposure, and crucially, whether that hedging effectiveness depends on the macroeconomic state.

Swap spreads—the difference between fixed rates on interest rate swaps and Treasury yields of similar maturity—are among the most actively traded relative-value measures in fixed income markets. Practitioners often view swap spread positions, particularly receive-fixed swap positions combined with long Treasury holdings, as providing protection during credit sell-offs. The intuition is that in periods of systemic stress, Treasury yields fall due to flight-to-quality demand while swap rates decline less, causing swap spreads to widen. If credit losses coincide with such spread widening, swap spread positions may serve as effective hedges.

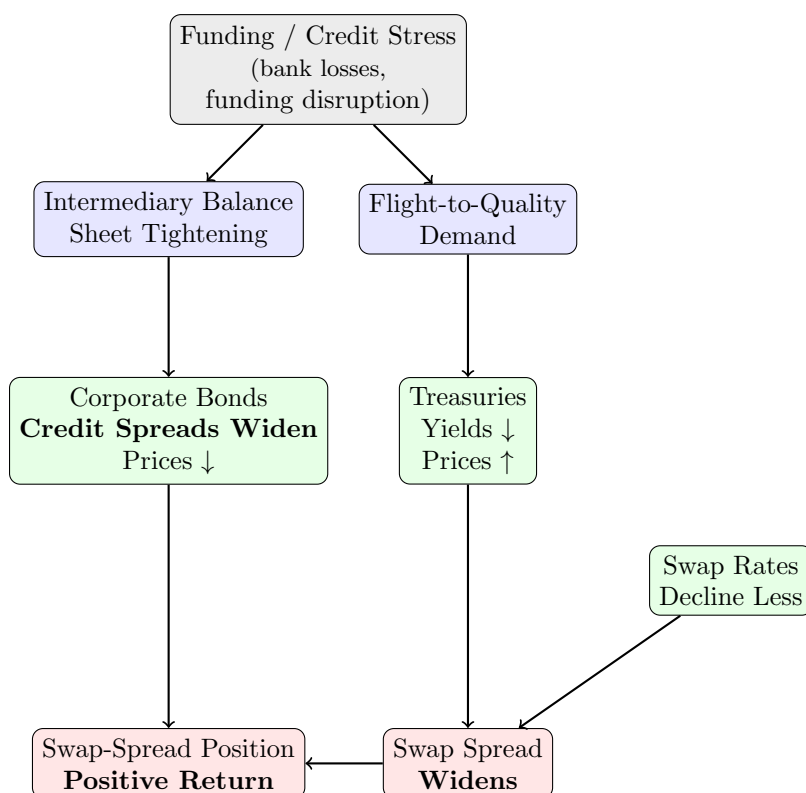


Figure 1: **Traditional View: Swap Spreads as a Hedge for Credit Exposure.** A funding or credit stress shock tightens intermediary balance sheets and triggers flight-to-quality demand for Treasuries. Treasury yields fall while swap rates decline less due to balance-sheet costs, causing swap spreads to widen. Because credit spreads widen and corporate bond prices fall simultaneously, swap-spread positions generate positive returns when credit suffers losses, creating a natural hedge under funding-stress conditions.

However, not all credit sell-offs are alike. Some arise from financial-sector distress and funding market disruptions, while others stem from inflationary shocks or monetary tightening that push risk-free rates higher. In these environments, both Treasuries and swaps may decline in value even as credit spreads widen. This raises a fundamental question: do swap spreads hedge credit exposure

in general, or only under specific macroeconomic conditions?

This paper provides both a theoretical framework and empirical evidence showing that the hedging role of swap spreads is state-dependent. We develop a simple intermediary-based asset pricing model in which dealer balance sheet constraints drive deviations between Treasury yields and swap rates. In this framework, swap spreads widen when intermediary constraints tighten and safe-asset demand rises. Because such tightening also amplifies credit risk premia, swap spread positions hedge credit exposure when shocks originate from funding and balance sheet stress. By contrast, when shocks are driven by inflation or monetary policy tightening, risk-free rates rise and both legs of the swap spread position may incur losses, causing the hedge to fail.

The model yields three testable predictions. First, swap spread positions should deliver positive returns during credit drawdowns associated with funding stress and flight-to-quality dynamics. Second, the hedge effectiveness should weaken or reverse during inflation-driven or rate-driven credit sell-offs. Third, the strength of the hedge should be increasing in measures of intermediary constraint tightness, such as dealer leverage constraints or funding market stress indicators.

We test these predictions using portfolios that replicate swap spread positions across maturities and measure their ability to offset losses in corporate bond and credit index portfolios. We document that swap spreads provide meaningful protection during episodes such as the Global Financial Crisis, the European sovereign crisis, and the COVID-19 market disruption, all characterized by sharp increases in safe-asset demand and intermediary balance sheet strain. In contrast, during episodes such as the 2013 Taper Tantrum and the 2022 inflation shock, swap spread positions fail to hedge and may even exacerbate losses.

These findings contribute to three strands of literature. First, we extend the literature on swap spread determinants by linking intermediary balance sheet frictions not only to pricing but also to the effectiveness of hedging strategies. Second, we connect to the growing literature on intermediary asset pricing by showing that intermediary constraints shape the covariance structure between asset classes and thus the performance of macro hedges. Third, we contribute to the literature on safe assets and flight-to-quality dynamics by demonstrating that the protective role of “safe” relative-value trades depends critically on the nature of the underlying macro shock.

Our results have practical implications for risk management and portfolio construction. Investors frequently rely on historical correlations to design hedging strategies, implicitly assuming stability across states. We show that such relationships are contingent on the source of economic stress and the behavior of intermediaries. Understanding when hedges work—and when they fail—requires modeling the interaction between balance sheet constraints, safe asset demand, and credit risk.

The remainder of the paper proceeds as follows. Section 5 presents the model and derives theoretical predictions. Section ?? describes the data and construction of swap spread and credit portfolios. Section ?? evaluates baseline hedging effectiveness. Section ?? examines state dependence across macroeconomic regimes. Section ?? links hedge effectiveness to intermediary balance sheet measures. Section ?? discusses robustness and alternative explanations. Section ?? concludes.

2 Literature Review

This paper relates to five broad literatures: (i) the determinants of swap spreads and the pricing of swaps relative to Treasuries; (ii) intermediary-based asset pricing and balance-sheet constraints; (iii) safe assets, convenience yields, and flight-to-quality dynamics; (iv) funding liquidity and short-

term funding stress measurement; and (v) shock identification and state dependence in fixed-income and credit comovement.

2.1 Swap spreads: pricing between swaps and Treasuries

A central object in this paper is the swap spread—the difference between the par fixed rate on an interest rate swap and the yield on a Treasury of similar maturity. Early work emphasizes that swap spreads are not mechanically pinned down by a frictionless no-arbitrage relationship because swaps embed counterparty credit risk, differential funding and collateral conventions, and market segmentation across swap and cash Treasury markets. Classic models study swap valuation and the link between swap rates and credit risk, showing that default and counterparty risk can generate a positive wedge between swap rates and risk-free benchmarks (Duffie and Huang, 1996; Cossin, 1997). Related theoretical work highlights how swap spreads can be influenced by hedging demand and bond supply, and can be sensitive to the relative liquidity of the benchmark Treasury security versus the swap market (Grinblatt, 1994).

Beyond counterparty risk, subsequent evidence stresses that swap spreads reflect time-varying liquidity premia and funding conditions, not only expected default. In this spirit, Liu (2002) provides a structured analysis of swap spread movements emphasizing that the Treasury–swap wedge is strongly affected by market frictions (including liquidity and supply effects) and can deviate persistently from simple reduced-form credit-only explanations. This perspective is especially relevant for post-crisis episodes when swap spreads became unusually low or even negative at long maturities, a phenomenon difficult to rationalize with counterparty credit risk alone.

Recent work attributes unusual swap spread behavior to the interaction between hedging demand, Treasury market pricing, and dealer balance-sheet constraints. In particular, Klingler and Sundaresan (2019) argues that long-maturity swap spreads can turn negative when strong receiving-fixed demand meets binding dealer balance-sheet constraints, so that intermediaries require compensation to warehouse swap-related exposures while Treasury yields embed an elevated convenience yield. This mechanism aligns closely with the present paper’s emphasis that the swap–Treasury wedge depends on intermediary capacity and safe-asset demand, and motivates our state-contingent approach to hedging: swap spreads can widen sharply in funding crises (supporting hedging), yet behave very differently when macro shocks primarily operate through discount rates.

2.2 Intermediary-based asset pricing and balance-sheet constraints

Our model builds on intermediary-based asset pricing, where financial intermediaries’ risk-bearing capacity, leverage constraints, and value-at-risk limits shape equilibrium risk premia and cross-asset comovement. A key implication of this literature is that when intermediary constraints tighten, assets requiring balance-sheet intermediation command higher expected returns, and relative value relationships become state dependent. Brunnermeier and Pedersen (2009) formalize the amplification between funding liquidity and market liquidity: shocks that impair funding conditions reduce intermediaries’ ability to provide liquidity, raising required compensation and generating joint dislocations across markets.

A complementary body of work emphasizes leverage cycles and the role of dealer balance sheets in propagating shocks. In these frameworks, intermediaries are not passive arbitrageurs; instead, their balance sheets are state variables that determine the pricing of risk and the strength of hedging

relationships across asset classes. This logic is central to our empirical prediction that swap-spread hedges are most effective precisely when intermediaries are constrained—because both credit risk premia and swap spread compensation are jointly tied to the same balance-sheet scarcity.

Recent research on Treasury market functioning underscores that dealer capacity is quantitatively important for prices and liquidity in core fixed-income markets. For example, [Duffie \(2025\)](#) surveys the institutional structure of the U.S. Treasury market and highlights the central role of dealers and balance-sheet constraints in Treasury market liquidity and stress transmission. This is crucial for our setting because swap spreads are defined relative to Treasuries; when Treasury convenience yields rise sharply (safe-asset demand) and dealers’ ability to intermediate cash-versus-derivatives positions is impaired, the swap–Treasury wedge and its covariance with credit can change abruptly.

The March 2020 episode provides vivid evidence of this intermediary channel: Treasury markets experienced severe strains despite Treasuries’ safe-asset status, consistent with binding intermediation constraints and large balance-sheet-intensive positioning flows. [Haddad, Moreira, and Muir \(2021\)](#) document dislocations and the role of dealer intermediation capacity in that episode, reinforcing the idea that safe-asset pricing and relative-value wedges (including swap spreads) can be deeply state contingent.

2.3 Safe assets, convenience yields, and flight-to-quality

A core mechanism in our paper is that funding stress triggers flight-to-quality: Treasuries become especially valuable as safe and liquid collateral, depressing Treasury yields relative to other instruments and contributing to swap spread widening when swap rates decline less than Treasury yields. This channel connects directly to the literature on safe assets and convenience yields. [Krishnamurthy and Vissing-Jorgensen \(2012\)](#) show that Treasuries can carry a convenience yield that varies over time with safe-asset demand, and that changes in the supply of safe assets can move Treasury yields beyond what is implied by standard term-premium decompositions. Related work similarly emphasizes that “money-like” or “safe” characteristics command premia in equilibrium and vary with macro-financial conditions, affecting price wedges between safe and near-safe claims ([Gorton, Lewellen, and Metrick, 2012](#)).

These insights are essential for interpreting swap spreads as hedges. If the Treasury leg of a swap-spread position earns a large convenience yield precisely in funding-stress states, then a DV01-neutral swap-spread package can produce positive returns when credit sells off. But if shocks are instead inflationary or policy-driven—raising expected short rates and discount rates broadly—Treasury prices may fall and the “safe-asset premium” may not rise enough to generate the same hedge benefit. Our empirical design therefore treats the source of stress (funding vs. inflation/rate shocks) as the key conditioning variable for whether the convenience-yield channel is activated.

2.4 Funding liquidity measures and stress states

Empirically implementing the intermediary channel requires measuring funding stress and balance-sheet tightness. A large literature uses money-market and interbank spreads to proxy funding liquidity conditions. For example, the FRA–OIS spread and related measures rose sharply during the global financial crisis and other funding disruptions, serving as high-frequency indicators of stress in unsecured or term funding markets and bank balance sheets. This measurement tradition

motivates our use of funding-spread innovations (including FRA–OIS, repo-related measures, and dealer leverage proxies) to construct empirical counterparts to funding-stress shocks.

An important methodological implication is that funding-stress proxies must be separated from movements in policy expectations and inflation news. In practice, funding spreads can move for multiple reasons, so our approach aligns with the broader asset-pricing practice of combining multiple funding indicators and extracting a common component, while separately measuring discount-rate shocks using high-frequency monetary and inflation surprises.

2.5 Shock identification and state dependence in fixed income and credit

Our state-dependent design also builds on the high-frequency identification literature used to isolate monetary policy shocks and inflation surprises from endogenous macroeconomic responses. [Kuttner \(2001\)](#) develops a widely used approach based on federal funds futures to measure unexpected policy actions, and the broader high-frequency framework is extended and refined in later work for identifying monetary policy surprises and their term-structure impacts. These methods motivate our construction of inflation/monetary shocks from narrow-window futures-based surprises and breakeven/real-yield innovations, which we use to define rate-driven states distinct from funding-driven states.

By combining these identification tools with intermediary-constraint proxies, our paper contributes a specific and testable implication: hedging relationships in fixed income that are often treated as “structural” can fail when the dominant shock shifts from balance-sheet scarcity (funding stress) to discount-rate news (inflation/monetary tightening). This state-contingent view complements a growing set of findings across macro-finance showing that cross-asset correlations and hedging effectiveness are not stable objects but vary with the underlying source of risk and the constraint set faced by intermediaries.

2.6 Contribution relative to existing work

Relative to the swap spread literature, our contribution is to shift the focus from unconditional determinants of swap spreads to the *state-contingent covariance* between swap-spread returns and credit returns—i.e., when and why swap spreads hedge credit exposure. Relative to intermediary asset pricing, we contribute evidence that intermediary constraints do not merely shift average premia, but also reshape the hedge effectiveness of widely used relative-value positions. Relative to the safe-asset literature, we show that Treasury convenience-yield dynamics help explain why swap-spread hedges work in funding crises but break down in inflation/rate-driven drawdowns. Finally, relative to the identification literature, we provide an empirically implementable mapping from structural shocks (funding vs. inflation/monetary) to observed state dependence in hedging outcomes.

3 Data and State Classification

This section describes the data used in the empirical analysis and the construction of macro-financial states that discipline the model’s predictions. The objective is to distinguish periods in which credit losses are driven by intermediary funding stress from periods driven primarily by inflationary or monetary tightening forces.

3.1 Swap Spreads

For maturity $\tau \in \{2, 5, 10, 30\}$ years, the swap spread is defined as

$$SS_t^\tau \equiv S_t^\tau - y_t^{T,\tau}, \quad (1)$$

where S_t^τ is the fixed rate on a plain-vanilla interest rate swap of maturity τ and $y_t^{T,\tau}$ is the constant-maturity U.S. Treasury yield of the same maturity. All series are observed at daily frequency and expressed in basis points.

To isolate relative pricing between swaps and Treasuries, we focus on changes in swap spreads. For maturity τ , the swap-spread return is defined as

$$R_t^{SS,\tau} \equiv \Delta SS_t^\tau = \Delta \left(S_t^\tau - y_t^{T,\tau} \right). \quad (2)$$

This object corresponds to a DV01-neutral position consisting of receiving fixed on a τ -year interest rate swap and shorting a matched-maturity Treasury. By construction, it removes parallel shifts in the risk-free yield curve and isolates movements in the swap–Treasury wedge.¹

3.2 Credit Exposure Measures

Credit exposure is measured using three complementary series:

1. **Investment-grade and high-yield credit spreads:** daily changes in option-adjusted spreads (OAS);
2. **CDX index spreads:** daily changes in CDX IG and CDX HY spreads;
3. **Credit excess returns:** duration-adjusted excess returns on corporate bond indices.

All credit variables are expressed as either spread changes (in basis points) or excess returns, ensuring comparability across specifications.

3.3 Market Volatility and Risk Conditions

Because direct measures of dealer balance-sheet stress (e.g., repo specialness, dealer leverage) are unavailable, we rely on market-based proxies that capture broad financial stress and uncertainty:

- **MOVE index:** measures implied volatility in the U.S. Treasury market and captures uncertainty related to rates, inflation, and monetary policy;
- **VIX index:** measures implied volatility in equity markets and captures broad risk-off conditions.

Both indices are observed at daily frequency.

¹Equivalently, one can scale swap-spread changes by maturity-specific DV01s. Because the analysis focuses on relative pricing and state-dependent comovement rather than level effects, all results are robust to such scaling.

3.4 State Classification

The model distinguishes two structural regimes: funding-stress states, in which intermediary balance sheets are constrained and flight-to-quality dynamics dominate, and inflation/monetary-tightening states, in which discount-rate shocks dominate without acute funding stress.

We classify states using observable volatility and rate dynamics.

Funding-Stress State. A day t is classified as a funding-stress state if:

$$\text{VIX}_t > \text{p75}(\text{VIX}) \quad \text{and} \quad \Delta y_t^{T,10} < 0, \quad (3)$$

where $\text{p75}(\cdot)$ denotes the 75th percentile computed over the full sample. This definition captures periods of elevated risk aversion combined with falling Treasury yields, consistent with flight-to-quality behavior and constrained intermediary balance sheets.

Inflation / Monetary-Tightening State. A day t is classified as an inflation or rate-driven state if:

$$\text{MOVE}_t > \text{p75}(\text{MOVE}) \quad \text{and} \quad \Delta y_t^{T,10} > 0. \quad (4)$$

This definition isolates periods of elevated interest-rate uncertainty accompanied by rising Treasury yields, consistent with inflationary or monetary-tightening shocks rather than funding stress.

Neutral State. Days that satisfy neither condition are classified as neutral and are excluded from state-dependent regressions unless otherwise stated.

3.5 Interpretation

The state definitions are designed to map directly to the model’s structural shocks. Funding-stress states correspond to realizations of dZ_t^F that generate safe-asset demand and swap-spread widening, while inflation states correspond to realizations of dZ_t^I that raise discount rates and weaken the swap-spread hedge.

By conditioning on observable volatility and yield movements, this classification isolates the macroeconomic origin of credit stress rather than relying on unconditional correlations. All state-dependent results are robust to alternative percentile thresholds (70th and 80th percentiles) and to weekly aggregation.

4 Identification Strategy

The core empirical question is whether swap spreads hedge credit exposure specifically through intermediary balance-sheet channels, rather than through general interest-rate or macroeconomic comovement. Given the absence of direct measures of dealer balance sheets or funding markets at daily frequency, our identification strategy relies on *state-based classification* rather than direct shock extraction. We distinguish periods in which market outcomes are consistent with funding-stress dynamics from periods dominated by inflationary or monetary tightening forces, and test whether hedging effectiveness differs systematically across these states.

4.1 Two Distinct Structural Shock Channels

The model features two conceptually distinct sources of aggregate risk:

1. **Funding-Stress (Intermediary) Shocks:** shocks that tighten intermediary balance-sheet constraints and reduce risk-bearing capacity. These shocks generate flight-to-quality demand for Treasuries, widen swap spreads through intermediation premia, and increase credit risk premia.
2. **Inflation / Rate Shocks:** shocks that raise expected future short rates or inflation expectations. These shocks move the risk-free yield curve and discount rates broadly, without necessarily inducing acute balance-sheet stress.

The key empirical challenge is to distinguish periods in which observed credit losses and swap-spread movements are driven primarily by the first channel from periods dominated by the second.

4.2 State-Based Identification Using Market Observables

Rather than attempting to extract latent structural shocks directly, we classify economic states using observable market variables that reflect the *realized effects* of different shocks.

Specifically, we use:

- **Equity market volatility (VIX)** as a proxy for broad risk aversion and stress conditions;
- **Treasury market volatility (MOVE)** as a proxy for uncertainty related to interest rates, inflation, and monetary policy;
- **Changes in long-maturity Treasury yields** to distinguish flight-to-quality episodes from rate-tightening episodes.

These variables are observed at daily frequency and allow us to classify periods according to the dominant macro-financial force affecting asset prices.

4.3 Funding-Stress States

A day is classified as a *funding-stress state* when:

$$\text{VIX}_t > \text{p75}(\text{VIX}) \quad \text{and} \quad \Delta y_t^{T,10} < 0. \quad (5)$$

This combination captures periods of elevated risk aversion coinciding with falling Treasury yields, consistent with flight-to-quality behavior and constrained intermediary balance sheets. In the model, these states correspond to realizations of the funding-stress shock dZ_t^F , which simultaneously widen credit spreads and swap spreads.

4.4 Inflation / Rate-Shock States

A day is classified as an *inflation or rate-driven state* when:

$$\text{MOVE}_t > \text{p75}(\text{MOVE}) \quad \text{and} \quad \Delta y_t^{T,10} > 0. \quad (6)$$

These periods are characterized by heightened uncertainty about interest rates and rising Treasury yields, consistent with inflationary or monetary tightening shocks rather than funding stress. In the model, these states correspond to realizations of dZ_t^I , which primarily affect discount rates and weaken the swap-spread hedge.

4.5 Mapping to the Structural Model

The state classification provides a reduced-form mapping to the model’s structural shocks:

$$\text{Funding-stress states} \longrightarrow dZ_t^F \quad (\text{tightening intermediary constraints})$$

$$\text{Inflation/rate states} \longrightarrow dZ_t^I \quad (\text{discount-rate shocks})$$

The model predicts that only funding-stress states generate simultaneous swap-spread widening and credit losses, implying positive hedging performance of swap-spread positions. Inflation or rate-driven states, by contrast, need not produce this comovement and may lead to hedge breakdown.

4.6 Interpretation and Identification Assumptions

Our identification strategy does not require direct observation of intermediary balance sheets. Instead, it exploits the fact that different structural shocks generate distinct and observable price patterns across markets. By conditioning on volatility measures and the sign of Treasury yield movements, we isolate periods in which balance-sheet constraints are likely to be the dominant pricing force.

This approach aligns with a large macro–finance literature that identifies economic mechanisms through state-dependent comovement rather than direct measurement of latent constraints. The resulting estimates should therefore be interpreted as conditional correlations that reflect different underlying structural environments, rather than causal responses to directly observed shocks.

Taken together, this strategy allows us to attribute variation in swap-spread hedge effectiveness to intermediary balance-sheet conditions versus discount-rate shocks in a manner consistent with both the model and the available data.

5 A Model of State-Dependent Hedging

5.1 Discussion: Why State-Dependent Intermediation?

This paper studies hedging in an environment where the effectiveness of commonly used financial hedges depends not only on the magnitude of shocks, but on their economic origin. Empirically, swap-spread positions are often viewed as reliable hedges against credit risk during episodes of funding stress, yet they perform poorly during inflation-driven or monetary-tightening episodes. The central question is therefore not whether swap spreads hedge credit risk on average, but when and why they do so.

Our modeling choice is guided by a simple organizing principle: intermediary balance sheets are the key marginal pricing factor in some states of the world, but not in others. When funding conditions deteriorate, intermediary net worth declines, balance-sheet constraints bind, and assets

that rely on intermediation capacity—such as corporate bonds and interest rate swaps—command additional compensation. In these states, price movements across markets are tightly linked through the common scarcity of balance-sheet capacity. By contrast, monetary or inflation shocks primarily affect discount rates and expected policy paths without directly alleviating intermediary constraints. In such states, the balance-sheet channel weakens and traditional hedging relationships can break down.

To capture this mechanism parsimoniously, we model intermediation capacity using a single state variable, intermediary net worth. This variable governs the shadow cost of balance-sheet usage and enters asset pricing through state-dependent intermediation premia. The model deliberately abstracts from detailed institutional features of individual markets in order to highlight a unifying force: when balance-sheet scarcity is the marginal constraint, hedging relationships are strong; when it is not, they need not hold. This abstraction allows the model to generate sharp, testable predictions about the state dependence of swap spreads, credit spreads, and hedging effectiveness across different macro-financial regimes.

5.2 Environment

Time is continuous. Uncertainty is driven by two aggregate Brownian shocks:

- a **funding-stress shock** dZ_t^F that tightens intermediary balance sheet constraints;
- an **monetary/inflation shock** dZ_t^I that raises expected future short rates.

The short rate evolves as

$$dr_t = \mu_r dt + \sigma_r^I dZ_t^I, \quad (7)$$

so monetary shocks directly affect the risk-free rate, while funding shocks operate through intermediary constraints and asset risk premia.

There are three traded securities:

- **Treasuries:** default-free government bonds,
- **Swaps:** fixed-for-floating interest rate swaps,
- **Corporate bonds:** risky bonds with default intensity λ_t .

Let M_t denote the stochastic discount factor (SDF).

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5.3 Intermediaries

A representative intermediary intermediates swaps and corporate bonds and faces limited risk-bearing capacity due to leverage or value-at-risk constraints. Net worth evolves as

$$dW_t = \mu_W dt + \sigma_W dZ_t^F, \quad (8)$$

where $\sigma_W < 0$ implies funding stress reduces net worth and tightens constraints.

The intermediary must satisfy

$$\text{Risk Exposure}_t \leq \kappa W_t, \quad (9)$$

with $\kappa > 0$. Lower net worth increases the marginal cost of balance sheet usage.

5.4 Asset Pricing

We describe asset returns in excess-return form, which avoids sign ambiguity and directly characterizes shock exposures.

5.4.1 Treasuries

Treasuries provide safety and liquidity. Their excess return satisfies

$$dR_t^T = \beta_T^F dZ_t^F + \beta_T^I dZ_t^I, \quad (10)$$

where $\beta_T^F > 0$ captures flight-to-quality: Treasury prices rise (yields fall) in funding stress.

5.4.2 Corporate Bonds

Corporate bond excess returns satisfy

$$dR_t^C = \beta_C^F dZ_t^F + \beta_C^I dZ_t^I, \quad (11)$$

with $\beta_C^F < 0$, since funding stress widens credit spreads and lowers corporate bond prices.

Credit spreads decompose as

$$s_t = \lambda_t + \phi_t, \quad \phi_t = \phi_0 + \frac{\phi_1}{W_t}, \quad \phi_1 > 0, \quad (12)$$

so tighter constraints raise intermediary risk premia. We decompose credit spreads into a fundamental default component and an intermediation component. The latter reflects the compensation required by intermediaries to warehouse balance-sheet-intensive positions. Even in normal times, regulatory capital requirements and market frictions imply a positive baseline intermediation premium. When intermediary net worth declines, balance-sheet constraints bind more tightly and the marginal cost of intermediation rises. To capture these forces parsimoniously, we model the intermediation premium as

$$\phi_t = \phi_0 + \frac{\phi_1}{W_t}, \quad \phi_1 > 0, \quad (13)$$

where ϕ_0 captures baseline intermediation costs and the second term captures the state-dependent scarcity of intermediary balance-sheet capacity. This formulation implies that

$$\lim_{W_t \rightarrow \infty} \phi_t = \phi_0 \neq 0,$$

reflecting the presence of intermediation frictions even when balance sheets are unconstrained.

5.4.3 Swaps

Let S_t denote the par fixed rate on a plain-vanilla interest rate swap. We decompose the swap rate into the risk-free short-rate (for example a benchmark implied by the expected path of SOFR) and an intermediation component. This decomposition stems from the idea that the swap fixed rate equals the market's expectation of future risk-free short rates plus the price or compensation required to convince or induce constrained dealers to intermediate the swap:

$$S_t = r_t + \eta_t, \quad \eta_t = \eta_0 + \frac{\eta_1}{W_t}, \quad \eta_1 > 0, \quad (14)$$

where η_t captures the compensation required by intermediaries to warehouse swap-related exposures. Although swaps are zero-net-supply contracts, dealers must absorb duration risk, collateral and funding requirements, and regulatory capital costs. Even in normal times, these frictions imply a positive baseline intermediation premium η_0 . When intermediary net worth declines, balance-sheet constraints bind more tightly and the marginal cost of intermediation rises, increasing the compensation required to intermediate swaps. As a result, swap rates decline less than Treasury yields of comparable maturity during funding-stress episodes, generating time variation in swap spreads. The parameter η_0 captures baseline intermediation costs that persist even when dealer balance sheets are unconstrained.

Define a DV01-neutral swap-spread position whose excess return is

$$dR_t^{SS} = \beta_{SS}^F dZ_t^F + \beta_{SS}^I dZ_t^I, \quad (15)$$

with $\beta_{SS}^F > 0$: swap spreads widen during funding stress.

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5.5 Swap Spreads

The swap spread is

$$SS_t = S_t - y_t^T. \quad (16)$$

Because both η_t and the safe-asset premium rise when W_t falls,

$$\frac{\partial SS_t}{\partial W_t} < 0. \quad (17)$$

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5.6 Hedging Mechanism

Consider a credit position with return dR_t^C and a swap-spread position with return dR_t^{SS} . Under funding stress ($dZ_t^F > 0$):

$$dR_t^C < 0, \quad dR_t^{SS} > 0.$$

Thus swap-spread positions hedge funding-driven credit losses.

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5.7 Predictions

Proposition 1 (Funding Stress Hedge). *When intermediary net worth falls, swap spreads widen and credit risk premia rise, so swap-spread positions hedge credit exposure.*

Proposition 2 (Inflation Shock Breakdown). *Under inflation/monetary shocks, swap spreads do not reliably widen, so the hedge weakens or fails.*

Proposition 3 (Constraint Intensity). *Hedge effectiveness increases when intermediary constraints tighten.*

5.8 Testable Implications

The model delivers three testable implications. First, swap-spread positions hedge credit risk during funding-stress episodes, when intermediary balance-sheet constraints are the dominant pricing force. Second, under inflation or monetary-policy shocks, swap spreads do not reliably widen, and the hedging relationship weakens or fails. Third, hedge effectiveness is increasing in the tightness of intermediary constraints. In the empirical section, we test these implications using regime-dependent specifications that distinguish funding-stress and monetary regimes and examine how swap spreads and credit returns co-move across states.

6 Empirical Framework

This section implements empirical counterparts to the model’s state-dependent hedging mechanism. The model implies that credit returns and swap-spread returns are jointly driven by two conceptually distinct forces: (i) funding-stress episodes in which intermediary constraints and flight-to-quality dynamics dominate, and (ii) inflation/monetary episodes in which discount-rate news dominates. We do not attempt to recover latent structural shocks directly. Instead, we test the model’s implications using the observable state classification in Section 3.

Let R_t^C denote a credit return or spread-based credit loss measure (e.g., excess returns, credit spread changes, or CDX spread changes), and let $R_t^{SS,\tau}$ denote the swap-spread return at maturity τ , defined as

$$R_t^{SS,\tau} \equiv \Delta SS_t^\tau, \quad SS_t^\tau \equiv S_t^\tau - y_t^{T,\tau}.$$

As discussed in Section 3, $R_t^{SS,\tau}$ corresponds to the return on a DV01-neutral swap-spread package up to scaling by maturity-specific DV01s.

The model’s key sign prediction is that in funding-stress states swap spreads widen while credit performs poorly, whereas in inflation/rate-driven states swap spreads need not widen and hedging can weaken or fail. Accordingly, we expect a *negative* hedge coefficient in funding-stress states and a weak or non-negative coefficient in inflation/rate-driven states.

6.1 Baseline Hedge Regression

We begin with the unconditional hedge regression:

$$R_t^C = \alpha + \beta R_t^{SS,\tau} + u_t, \tag{18}$$

where $\beta < 0$ indicates that swap-spread positions tend to gain when credit loses value. Because different shock environments can generate offsetting comovement, the unconditional coefficient β may be weak or unstable, motivating state conditioning.

Throughout the empirical analysis, R_t^C denotes the excess return on a broad corporate bond index. Negative realizations of R_t^C therefore correspond to credit losses arising from spread widening, increases in default risk, or deteriorating funding conditions. Framing credit exposure in return space allows direct comparison with swap-spread portfolio returns and aligns the empirical analysis with the model’s return-based hedging predictions.

Table 1: Baseline Hedge Regression: Swap Spreads and Credit Returns (Pre-SOFR Sample)

Swap Spread Maturity	Dependent Variable: R_t^C (Corporate Bond Index)			
	2Y	5Y	10Y	20Y
ΔSS_t^τ	0.085 (0.055)	0.128** (0.059)	0.129 (0.107)	0.128 (0.121)
Constant	0.306*** (0.074)	0.306*** (0.074)	0.306*** (0.073)	0.306*** (0.073)
Observations	8,556	8,556	8,556	8,556
R^2	0.0008	0.0017	0.0013	0.0015
HAC Std. Errors	Yes	Yes	Yes	Yes
Sample	1997–2020	1997–2020	1997–2020	1997–2020

Notes: The table reports OLS estimates of the baseline hedge regression $\Delta R_t^C = \alpha + \beta \Delta SS_t^\tau + u_t$, where $\Delta CorpBondIndex_t^C$ is the excess return on a broad U.S. corporate bond index and ΔSS_t^τ denotes daily changes in the τ -year swap spread. Standard errors are Newey–West (HAC) with a fixed bandwidth of 11. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table 1 reports unconditional hedge regressions of corporate bond excess returns on swap-spread changes across maturities. Consistent with the model, unconditional hedge coefficients are small, unstable, and in several cases positive, indicating that swap spreads do not provide reliable hedging benefits on average. This reflects the coexistence of distinct macro-financial shock environments: funding-stress episodes, in which swap spreads widen and credit returns fall, and inflation- or rate-driven episodes, in which swap spreads fail to widen or may even narrow while credit returns deteriorate. The unconditional regression therefore averages across regimes with opposing covariance structures, motivating the state-dependent analysis that follows.

The unconditional results in Table 1 show little evidence that swap-spread positions hedge corporate bond returns on average. This is not a failure of the hedge, but a direct implication of the model: different macro-financial shocks generate offsetting comovement between swap spreads and credit returns. We therefore turn to state-dependent tests that condition on the source of credit stress.

6.2 State-Dependent Hedge Effectiveness

We estimate state-dependent hedge effectiveness by interacting swap-spread returns with the mutually exclusive states defined in Section 3:

$$R_t^C = \alpha + \beta_F \left(R_t^{SS,\tau} \cdot 1\{\text{Funding}_t\} \right) + \beta_I \left(R_t^{SS,\tau} \cdot 1\{\text{Inflation}_t\} \right) + u_t, \quad (19)$$

where $1\{\text{Funding}_t\}$ identifies funding-stress days (high VIX and falling Treasury yields) and $1\{\text{Inflation}_t\}$ identifies inflation/rate-driven days (high MOVE and rising Treasury yields).

The model predicts:

$$\beta_F < 0, \quad \beta_I \approx 0 \text{ or } \beta_I \geq 0,$$

so that swap spreads hedge credit primarily in funding-stress states, but the hedge weakens or fails in inflation/rate-driven states.

6.3 Continuous State-Intensity Tests

To assess whether hedge effectiveness varies smoothly with observable stress intensity (rather than only in discrete regimes), we estimate interaction specifications using the available volatility measures:

$$R_t^C = \alpha + \beta R_t^{SS,\tau} + \gamma_V(R_t^{SS,\tau} \times \widetilde{\text{VIX}}_t) + \gamma_M(R_t^{SS,\tau} \times \widetilde{\text{MOVE}}_t) + u_t, \quad (20)$$

where $\widetilde{\text{VIX}}_t$ and $\widetilde{\text{MOVE}}_t$ denote standardized (demeaned and scaled) versions of VIX and MOVE.

Consistent with the model, we expect the swap-spread hedge to strengthen when funding-stress conditions intensify (proxied by high VIX combined with flight-to-quality dynamics), while elevated rate uncertainty (high MOVE) need not strengthen the hedge:

$$\gamma_V < 0, \quad \gamma_M \approx 0 \text{ or } \gamma_M \geq 0.$$

6.4 Rate-Move Controls and Robustness

Because rate changes can mechanically affect both swap spreads and credit, we also consider specifications that control for Treasury yield movements:

$$R_t^C = \alpha + \beta R_t^{SS,\tau} + \delta \Delta y_t^{T,10} + u_t, \quad (21)$$

and we verify that the state-dependent results in (19) are robust to alternative percentile thresholds (70th/80th), to weekly aggregation, and to using different credit measures (OAS, CDX, and credit excess returns).

6.5 Interpretation

The empirical framework tests the model's central mechanism using only observable data: swap spreads, credit measures, VIX, MOVE, and Treasury yield changes. Evidence in favor of the model consists of (i) a materially negative hedge coefficient in funding-stress states ($\beta_F < 0$), (ii) weak or non-negative hedge coefficients in inflation/rate-driven states ($\beta_I \geq 0$), and (iii) stress-intensity interactions indicating stronger hedging when funding-stress conditions intensify.

Economically, swap spreads hedge credit risk not because they hedge default per se, but because both swap-spread returns and corporate bond returns load with opposite sign on funding-stress states that tighten intermediary balance sheets.

7 Appendix for Model Foundations and Proofs

This appendix formalizes the stochastic discount factor (SDF) economy underlying the model and provides proofs of Propositions 1–3.

7.1 A.1 State Variables

Uncertainty is driven by two Brownian shocks

$$dZ_t^F \quad (\text{funding stress}), \quad dZ_t^I \quad (\text{inflation/monetary}),$$

with $\mathbb{E}[dZ_t^j] = 0$ and $\mathbb{E}[dZ_t^j dZ_t^k] = \rho_{jk} dt$.

Intermediary net worth evolves as

$$dW_t = \mu_W dt + \sigma_W dZ_t^F, \quad \sigma_W < 0, \quad (22)$$

so positive funding stress reduces dealer balance-sheet capacity.

7.2 A.2 Stochastic Discount Factor

Asset prices are determined by the intermediary SDF:

$$\frac{dM_t}{M_t} = -r_t dt - \lambda_F dZ_t^F - \lambda_I dZ_t^I, \quad (23)$$

where $\lambda_F > 0$ increases when intermediary constraints tighten. A convenient reduced form is

$$\lambda_F = \lambda_0 + \frac{\lambda_1}{W_t}, \quad \lambda_1 > 0.$$

No-arbitrage implies for any asset j :

$$\mathbb{E}_t[dR_t^j] = (\lambda_F \beta_j^F + \lambda_I \beta_j^I) dt. \quad (24)$$

7.3 A.3 Asset Return Exposures

Let excess returns satisfy

$$dR_t^j = \beta_j^F dZ_t^F + \beta_j^I dZ_t^I. \quad (25)$$

Key sign restrictions implied by the model:

Asset	β^F (Funding Stress)	β^I (Inflation)
Treasuries	< 0 (flight to quality)	< 0 (duration loss)
Corporate Bonds	> 0 (spread widening)	> 0 (rate-driven losses)
Swap Spread Portfolio	> 0 (spread widening)	≤ 0 (weak or negative)

The swap-spread portfolio corresponds to receiving fixed in swaps and hedging Treasury duration, isolating relative pricing between swaps and Treasuries.

7.4 A.4 Proof of Proposition 1

Proof. Under funding stress, asset returns load on dZ_t^F :

$$dR_t^C = \beta_C^F dZ_t^F + \dots, \quad dR_t^{SS} = \beta_{SS}^F dZ_t^F + \dots$$

with $\beta_C^F > 0$ and $\beta_{SS}^F > 0$. Thus

$$\text{Cov}(dR_t^{SS}, dR_t^C) = \beta_{SS}^F \beta_C^F dt > 0.$$

Since funding stress corresponds to negative corporate bond returns, a position with positive β_{SS}^F delivers gains when credit loses value.

The hedge ratio $\theta^* = -\beta_C^F / \beta_{SS}^F$ eliminates funding-risk exposure:

$$dR_t^C + \theta^* dR_t^{SS} = 0 \cdot dZ_t^F + \dots$$

Hence swap spreads hedge funding-driven credit losses. □

7.5 A.5 Proof of Proposition 2

Proof. Under inflation shocks:

$$dR_t^C = \beta_C^I dZ_t^I, \quad dR_t^{SS} = \beta_{SS}^I dZ_t^I.$$

With $\beta_C^I > 0$ and $\beta_{SS}^I \leq 0$,

$$\text{Cov}(dR_t^{SS}, dR_t^C \mid dZ_t^I) = \beta_{SS}^I \beta_C^I dt \leq 0.$$

Swap spreads therefore do not offset inflation-driven credit losses and may move in the same direction as credit. □

7.6 A.6 Proof of Proposition 3

Proof. From (24), expected compensation per unit of funding-risk exposure is λ_F . Since

$$\frac{\partial \lambda_F}{\partial W_t} = -\frac{\lambda_1}{W_t^2} < 0,$$

funding-risk pricing intensifies as intermediary net worth declines.

Assets with higher β^F become more sensitive to funding shocks in equilibrium pricing, strengthening the covariance structure between swap spreads and credit. Therefore hedge effectiveness increases as constraints tighten. □

7.7 A.7 Proof of State-Dependent Regression Beta

Proof. The OLS coefficient from

$$dR_t^C = \alpha + \beta dR_t^{SS} + \varepsilon_t$$

equals

$$\beta = \frac{\text{Cov}(dR_t^C, dR_t^{SS})}{\text{Var}(dR_t^{SS})}.$$

Using (25),

$$\begin{aligned}\text{Cov}(dR_t^C, dR_t^{SS}) &= (\beta_C^F \beta_{SS}^F + \beta_C^I \beta_{SS}^I) dt, \\ \text{Var}(dR_t^{SS}) &= ((\beta_{SS}^F)^2 + (\beta_{SS}^I)^2) dt.\end{aligned}$$

Thus

$$\beta = \frac{\beta_C^F \beta_{SS}^F + \beta_C^I \beta_{SS}^I}{(\beta_{SS}^F)^2 + (\beta_{SS}^I)^2}.$$

The sign depends on whether funding or inflation components dominate. □

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